

Curriculum Vitae

Ir. Dr. Zhiheng ZHAO (Billy Zhao)

A BASIC INFORMATION

A.1. PERSONAL DETAIL

Full name and title	Ir. Dr. Zhiheng ZHAO (<i>Billy Zhao</i>)
Gender	Male
Nationality	Hong Kong S.A.R, China
Contact address	CF411, Core C, The Hong Kong Polytechnic University, Hung Hom, Kowloon, Hong Kong
Tel	852 27666630 (Office); 852 54055009 (Mobile)
Email	zhiheng.zhao@polyu.edu.hk
URL	https://www.polyu.edu.hk/ise/people/academic-staff/zhiheng-zhao/

A.2. ACADEMIC QUALIFICATIONS

Date	Degree in discipline
Dec, 2017	Ph.D. in Logistics Engineering, The University of Hong Kong, Hong Kong
Jun, 2013	M.Sc. in Industrial Engineering and Logistics Management, The University of Hong Kong, Hong Kong
Jun, 2012	Bachelor of Business Administration, Zhengzhou University, China

A.3. WORK EXPERIENCE

Date	Position held
Mar 2026 – Present	<i>Assistant Professor</i> , Department of Industrial & Systems Engineering, The Hong Kong Polytechnic University, Hong Kong
Feb 2023 – Mar 2026	<i>Research Assistant Professor</i> , Department of Industrial & Systems Engineering, The Hong Kong Polytechnic University, Hong Kong
Jul 2021 – Feb 2023	<i>Associate Professor (Tenured)</i> , School of Internet of Things, Nanjing University of Posts and Telecommunications, China
Mar 2018 – Jun 2021	<i>Lecturer</i> , School of Internet of Things, Nanjing University of Posts and Telecommunications, China
Oct 2017 – Mar 2018	<i>Postdoctoral Fellow</i> , Department of Industrial and Manufacturing Systems Engineering, The University of Hong Kong, Hong Kong

A.4. HONOURS AND AWARDS

	Date of Award	Title of award	Name of the awarding institution/body
*	2025	Best Report First Prize, ‘AI agent-enhanced operation management for Cyber-physical Internet-enabled fixed-position operation islands’, Chen, Q., Zhao, Z., Huang, G.Q.	Annual Conference of Logistics and Supply Chain Innovation Alliance of Guangdong-Hong Kong-Macao Greater Bay Area-Empowering Innovation in Supply Chain and Logistics Management with Artificial Intelligence
*	2025	Physical Internet Ambassador	European Union
*	2025	Best Conference Paper Award 2 nd Prize, ‘Look-around reasoning-enabled resource allocation in production logistics within Industry 4.0’, Li, J., Zhao, Z.	NSFC-UGC Frontiers of Digital Twins in Intelligent Manufacturing and Smart Cities
*	2025	Best Healthcare Automation Paper Award - Finalist, ‘Out-of-distribution modular hospital fit-out scheduling via memory-augmented deep reinforcement learning’, Han, Y., Sun, K., Zhao, Z., Huang, G.Q.,	IEEE Robotics and Automation Society, IEEE International Conference on Automation Science and Engineering- CASE 2025
*	2025	Physical Internet π -Builder Award 2025	International Physical Internet Conference, Hong Kong, China
*	2024	Best Digital Manufacturing and Human-Centered Automation Paper in Application, ‘Internet of Everything and Digital Twin enabled Service Platform for Cold Chain Logistics’, Wu, W., Shen, L., Zhao, Z., Harish, A., Zhong	IEEE Robotics & Automation Society, Technical Committee on Digital Manufacturing and Human-Centered Automation

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		R., Huang, G.Q.	
*	2024	Best Paper Award ‘室内定位技术驱动的车间物流智能管控方法及应用研究’, 吴畏, 赵之恒, 李聪波, 黄国全	2024 年中国工业工程年会暨第十三届工业工程企业应用与实践高峰论坛
*	2024	Highest External Competitive Research Grant Award, Bronze Award 2023/24	ISE, PolyU
*	2024	Award for High SFQ Score 2023/24	ISE, PolyU
*	2024	Best Poster Award, ‘DT-PosFormer: A Digital Twin-enabled Transformer Network for Precise Pose Estimation and Trajectory Prediction of MiC Modules’, Han, Y., & Zhao, Z.	10 th International Physical Internet Conference, Savannah, GA USA
*	2023	Outstanding Paper Award, ‘Logic-based Benders Decomposition for Order Acceptance and Scheduling in Distributed Manufacturing’, Chen, J., Ma, W, Ye, X., Zhao, Z.	Annual International Conference for Chinese Scholars in Industrial Engineering & 13th Forum for Council of Industrial Engineering and Logistic Management Department Heads
*	2023	Best Paper First Prize, ‘On Spatial-temporal Traceability and Relativity for Industry 4.0 Applications’, Zhao, Z., Zhang, M., Chen, Q., Huang, G. Q., Wu, W.	Annual International Conference for Digital Intelligence Logistics and Supply Chain Management
	2022	Young Teachers Excellent Teaching Award	NJUPT
	2020	First-class Award, Young Talent of Teaching Competition	NJUPT
	2019	Entrepreneurship and Innovation Talent Award 江蘇省“雙創人才”	Jiangsu Government, China
	2016	Best Paper Award, “Optimization based transportation service trading in B2B e-commerce logistics”, Zhang, M., Huang, G. Q., Xu, S., & Zhao, Z.	The 7th Forum for Council of Industrial Engineering and Logistics Management Department Heads & The 5th Institute of Industrial and Systems Engineers Asia Conference.

**Awarded after joining PolyU on February 2023.*

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A.5. SUMMARY OF ACHIEVEMENTS

Research:

# of Journal Papers:	63	
# of Conference Papers:	22	
# of <i>Research Grant Council</i> Projects as PI:	1 GRF	\$1.17M
# of <i>Research Grant Council</i> Projects as Co-PI/Co-I:	2 (1 RIF as Co-PI, 1 TRS as Co-I)	\$1.40M
# of <i>National Science Foundation of China</i> Projects PI:	1 Youth Scientist	\$0.31M
# of <i>Policy Research Funding</i> Projects as PI:	1 PPRFS	\$0.49M
# of <i>Innovation and Technology Fund</i> Projects as PI:	1 PRP	\$4.24M
# of <i>Joint Laboratory</i> Established as PI:	1 on Industrialized Construction	\$2/10M (Received/MoU signed for 5 years)
# of <i>Donations</i> Received as PI:	1 Donation	\$2.00M
# of <i>Other External Competitive Research</i> Projects as PI:	6	\$0.68M
# of <i>Other Industrial Collaborative Research</i> Projects as PI:	12	\$7.01M
Total External Research Income as PI:		\$20.03M/28.03M

Teaching:

# of courses taught:	7 (4 courses since joining PolyU)
PolyU <i>eSFQ</i> Score:	4.6 (2023/24 S1), 4.3 (2024/25 S1), 4.9 (2024/25 S2)
Teaching ranking before PolyU:	1/34 (2018/19 S2), 8/28 (2019/20 S2), 1/34 (2020/21 S1)
# of Ph.D. (MPhil) under supervision:	14(2)
# of Research Associate/PDF under supervision:	10
# of Research/Project Assistant under supervision:	5

Service:

# of Guest Editors, Conference Program/Organization Chairs:	>10
# of Invited Talks and Seminars:	>6
# of Reviewed Journals	>30
# of Best Paper Awards	>6
Distinction:	Physical Internet Ambassador (European Union) Physical Internet π -Builder Award 2025 Highest External Competitive Research Grant Award 入選江蘇省雙創計劃 (省級人才稱號)

Knowledge Transfer and Entrepreneurship:

# of Patents/Software Copyrights	>10
Entrepreneurship	CEO of Gantan Limited (www.gantanmark.com)
Investment from Government	Hangzhou Qingshan Lake Program 2019 ¥4.00M Cyberport Incubation Program 2017 \$0.33M
Major Clients	Huawei Technologies Ltd, Hong Kong Air Cargo Terminals Ltd, Lenovo Group Ltd, DKSH Management Ltd, and Gree Electric Appliances. Inc, etc.

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B DETAILS ON RESEARCH

Billy has been particularly successful in securing substantial research funding from diverse sources, including RGC, NSFC, ITC, PPPFS, industrial sponsors and donations. Within less than three years at PolyU, he has attracted over HK\$22.69 million in external research income and established a joint laboratory. This significant level of funding has greatly intensified research activities and enhanced capacity, leading to the publication of nearly 60 high-quality (Grade A) journal papers.

B.1. RESEARCH INPUTS (GRANTS/CONTRACT/DONATION PROJECTS)

B.1.1. External Peer-reviewed Competitive Research Grants as “PI”

No.	Project Title and Number	Funding Scheme	Amount of Grants	Period	Status of Project
*10	A spatial-temporal ‘look-around’ reasoning driven out-of-distribution dispatching approach for production logistics synchronization (No. 15203025)	<i>RGC General Research Fund</i>	HK\$ 1,171,772	01/2026-12/2028	On-going
*9	Harnessing the Synergistic Effect of Construction Robotics and Modular Integrated Construction: Key Barriers, Strategic Applications, and Policy Tools (No. 2025.A6.293.25C)	<i>Public Policy Research Funding Scheme</i>	HK\$ 483,209.30	11/2025-11/2026	On-going
*8	Research of real-time synchronization method of production logistics based on digital twin spatial-temporal reasoning (No. 52305557)	<i>National Natural Science Foundation of China</i>	HK\$ 319,680	01/2024-12/2026	On-going
*7	Auto-calibrated Real-time Locating Service-based Production Logistics Synchronization System (No. PRP/038/24LI)	<i>Innovation and Technology Fund</i>	HK\$ 4,240,000	06/2024-03/2026	On-going
*6	Research of digital twin and dynamic spatial-temporal knowledge graph driven real-time production logistics resources allocation (No. 2024A1515011930)	<i>Natural Science Foundation of Guangdong Province</i>	HK\$ 161,880	01/2024-12/2026	On-going
*5	Digital twin spatial-temporal reasoning based smart discrete manufacturing system (No. IMETKF2024022)	<i>State Key Laboratory of Intelligent Manufacturing Equipment and Technology</i>	CNY ¥ 50,000	01/2024-12/2025	On-going
4	Digital twin and dynamic spatial-temporal knowledge graph for PL resource allocation (No. 2022M712394)	<i>China Postdoctoral Science Foundation</i>	CNY ¥ 50,000	09/2022-08/2024	Completed
3	nD Resource synchronization in manufacturing management platform (No. 201930562)	<i>Innovative and Entrepreneurial Talent Program of Jiangsu</i>	CNY ¥ 150,000	07/2019-07/2021	Completed
2	Ubiquitous indoor positioning of supply chain resources in IoT environment (No. BK20180749)	<i>Natural Science Foundation of Jiangsu Province</i>	CNY ¥ 200,000	07/2018-06/2021	Completed
1	nD production resource tracking in industrial park (No. RK016NLX19007)	<i>Nanjing Municipal Human Resources and Social Security Bureau</i>	CNY ¥ 30,000	07/2019-07/2021	Completed

**Awarded after joining PolyU on February 2023.*

B.1.2. Other Research/Teaching/Contract/Donations Grants as “PI”

No.	Project Title	Funding Scheme	Amount of Grants	Period	Status of Project
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*14	Development of Construction Robots Digitalization System for Industry Metaverse (No. P0052836)	Nanjing Zhuling Technology Co., Ltd.	HK\$ 200,000/10,000,000 (Received/MoU signed for 5 years)	08/2024-07/2026	On-going
*13	A Human-Centered Autonomous Production Logistics Framework Driven by Industrial Large Models (No. P0057754)	Donation	HK\$ 2,000,000	09/2025-08/2030	On-going
*12	Smart AoA Indoor Positioning for Cyber-physical Internet Enabled Production and Logistics (No. P0048335)	RGC Research Matching Grant Scheme-Yunli Tech Limited	HK\$ 142,500	10/2023-04/2024	Completed
*11	Map-Assisted Indoor Localization Service for Blockchain-Integrated Industrial Applications (No. P0056272)	Transcell Intelligent Machinery (Changzhou) Co.Ltd	HK\$ 1,000,000	05/2025-05/2027	On-going
*10	Development of ESG Decision Support System Based on Large Language Model (No. P0056231)	Urumqi Zhuofei Electronic Technology Co., Ltd	HK\$ 1,280,000	06/2025-05/2029	On-going
*9	Development of Smart Educational System via Curriculum-learned Large Language Models (No. P0056130)	Guangzhou Xiayue Education and Technology Investment Co., Ltd	HK\$ 500,000	05/2025-04/2027	On-going
*8	Auto-calibrated Real-time Locating Service-based Production Logistics Synchronization System 1 (No. P0055554)	Projects of RIAM	HK\$ 500,000	04/2025-04/2027	On-going
*7	Joint Laboratory on Industrialized Construction (No. P0055873)	Projects of RIAM	HK\$ 500,000	04/2025-04/2027	On-going
*6	Digital twin spatial-temporal reasoning based real-time synchronization for production logistics (No. P0045749)	Departmental General Research Fund	HK\$ 400,000	06/2023-06/2025	Completed
*5	On spatial-temporal traceability and relativity for Industry 4.0 application (No. P0046067)	Start-up Fund for RAPs under the Strategic Hiring Scheme	HK\$ 500,000	06/2023-06/2025	Completed
*4	Postdoc Matching Fund Scheme 23/24 4th round - Dr Yadong XU (No. P0052893)	Postdoc Matching Fund Scheme	HK\$ 364,567	08/2024-07/2026	On-going
3	Parking lot management of dangerous and chemical goods (No. KJ016HZ19052)	Patent Transfer Fund	CNY ¥ 100,000	07/2019-06/2022	Completed
2	Exploration on talent cultivation in Internet of Things (No. JG01620JX20)	Teaching Reform Project of NJUPT	CNY ¥ 5,000	01/2020-12/2020	Completed
1	Location tracking of supply chain resources (No. NY218125)	Start-up Fund of NJUPT	CNY ¥ 50,000	07/2018-06/2020	Completed

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B.1.3. Other Research/Teaching/Contract Grants as “Co-PI” or “Co-I”

No.	Project Title	Funding Scheme	Role	Amount of Grants (Allocated)	Period	Status of Project
*13	Industry 4.0 Smart Prefabrication Yard for Modular Construction Fitout (No. R7036-22)	RGC Research Impact Fund	Co-PI	HK\$5,335,607.87 (1,400,000)	05/2023-04/2028	On-going
*12	SynchronHub: Cyber-Physical Internet for Synchronizing Cross-Border Logistics Hubs in the Greater Bay Area (GBA) (No. T32-707/22-N)	RGC Theme-based Research Scheme	Co-I	HK\$ 39,196,000	01/2023-12/2027	On-going
*11	AiFashion: Multi-modal and Multi-dimensional Large Model based on Self-Trained Customer Digital-Twin for Fashion Smart Social Manufacturing (No. PRP/015/24TI)	Innovation and Technology Fund	Co-I	HK\$ 4,469,978	03/2024-04/2026	On-going

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*10	Smart Welding Island Service System (SWISS): Human-robot Collaboration and Synchronization (No. PRP/007/25LI)	<i>Innovation and Technology Fund</i>	Co-I	HK\$ 6,000,000	04/2025-03/2027	On-going
*9	Cyber-Physical Internet (CPI) for Collecting and Calculating Carbon Emissions of Urban Transport and Logistics (No. P0050805)	<i>Department General Research Fund</i>	Co-I	HK\$ 800,000	06/2024-05/2026	On-going
*8	Smart Fashion Social Manufacturing: Resource Gateways, Network Services, and Decision Protocols (No. P0052206)	<i>Intra-Faculty Interdisciplinary Projects</i>	Co-I	HK\$ 500,000	09/2024-08/2026	On-going
*7	The Impact of Digital Twin on the Service Quality and Productivity of the Hospitality Industry (No. P0046028)	<i>SHTM Department Earnings</i>	Co-I	HK\$ 119,999	06/2023-04/2026	On-going
*6	TouristESG: A blockchain-based tourist data sharing platform for ESG certification (No. P0051281)	<i>RCDTT Fund</i>	Co-I	HK\$ 500,000	12/2024-11/2026	On-going
*5	Meta-Box for Cyber-Physical-Internet (No. P0046336)	<i>Cainiao Supply Chain Hong Kong Co., Ltd</i>	Co-I	HK\$ \$300,000	08/2023-01/2025	Completed
4	IoT-enabled supply chain smart service platform for agricultural products (No. 19KJA180006)	<i>Major Natural Science Research Project of Higher Education of Jiangsu Province</i>	Co-I	CNY ¥ 300,000	07/2019-06/2023	Completed
3	Research and practice of the new engineering education model integrating industry and education with diverse collaborative cultivation (No.2021JSJG710)	<i>Teaching Reform Project of Higher Education of Jiangsu Province</i>	Co-I	CNY ¥ 10,000	01/2022-12/2023	Completed
2	ESG indicators-based E-commerce logistics cooperation strategy research with the carbon-neutral goal (No. BK20220382)	<i>Natural Science Foundation of Jiangsu Province</i>	Co-I	CNY ¥ 200,000	07/2022-06/2025	Completed
1	Research on Optimization Strategies of E-commerce Logistics Service Transaction Mechanisms Considering Demand Disruption (No.19KJB580016)	<i>Natural Science Research of the Jiangsu Higher Education Institutions of China</i>	Co-I	CNY ¥ 50,000	09/2019-08/2021	Completed

*Awarded after joining PolyU on February 2023.

B.2. RESEARCH OUTPUTS

B.2.1. Refereed journal papers – listed as first or corresponding author:

40. Lin, J., Zhang, M., Zhao, Z.*, Huang, G. Q. (2026). Optimal pricing strategy in live streaming sales with blockchain traceability. *Expert Systems With Applications*.
39. Li, J., Fan, J., **Zhao, Z.***, Wang, L., & Huang, G. Q. (2026). Cyber-physical computer in action: Generative recommendation via spatial-temporal subgraph reasoning for production logistics orchestration. *IEEE Transactions on Industrial Informatics*.
38. Yuan, M., Chen, Q., **Zhao, Z.***, Thürrer, M., Qu, T., & Huang, G. Q. (2026). Cyber-physical internet-enabled out-of-order execution considering workload for multi-stage scheduling in hybrid flow shop. *Journal of Manufacturing Systems*, 86, 82-97.
37. Sun, M., Li, J., Fan, J., Zhang, M., **Zhao, Z.***, & Huang, G. Q. (2026). A double-sided bundle auction mechanism for collaborative additive manufacturing. *Journal of Manufacturing Systems*, 85, 473-486.
36. Yuan, Y., Wang, Y., Lu, Z., Shen, F., **Zhao, Z.***, & Huang, G. Q. (2026). Causally-Aware Hierarchical Graph Neural Networks for Robust Trajectory Prediction. *IEEE Transactions on Vehicular Technology*.
35. Sun, K., Han, Y., **Zhao, Z.***, & Huang, G. Q. (2026). Vision-Guided Fashion Fine-Grained Attribute Editing via Semantic Segmentation and Disentangled Representation. *IEEE Transactions on Circuits and Systems for Video Technology*.

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34. Sun, K., Han, Y., **Zhao, Z.***, & Huang, G. Q. (OCT 2025). Enhancing Large Language Models for Fashion Smart Manufacturing via Dynamic Collaborative Routing-Based Retrieval Reranking. *IEEE Transactions on Cybernetics*. (IF: 10.5, Q1)
33. Lu, Z., **Zhao, Z.***, & Huang, G. Q. (JAN 2026). STAR: Spatial–Temporal Attention Reasoning model for dynamic logistics network routing in Cyber–Physical Internet. *Advanced Engineering Informatics*, 69, 103830. (IF: 9.9, Q1)
32. Wang, Y., **Zhao, Z.***, Han, Y., Cheng, Y., & Huang, G. Q. (DEC 2025). Cyber-physical internet enabled hierarchical attention network based reinforcement learning for order dispatch in fast fashion manufacturing. *Journal of Manufacturing Systems*, 83, 784–799. (IF: 14.2, Q1)
31. Li, J., **Zhao, Z.***, Yang, C., Huang, S., Lee, L., & Huang, G. Q. (NOV 2025). ChatSync: Large Language Model Enabled Spatial-Temporal Knowledge Reasoning for Production Logistics Synchronization. *IEEE Internet of Things Journal*. (IF 8.9, Q1)
30. Xu, Y., Ji, J. C., Sun, Y., Huang, S., **Zhao, Z.***, & Huang, G. Q. (OCT 2025). Self-supervised time–frequency feature alignment for process monitoring of cyber–physical CNC machines. *Journal of Manufacturing Systems*, 82, 1145–1157. (IF: 14.2, Q1)
29. Li, Z., Liu, J., Xu, Y., Lee, L. H., & **Zhao, Z.*** (NOV 2025). A computational study on the hydrodynamics of real-time pitch manipulation in dolphins during self-propulsion: The promoting effect of caudal fin under offset sinusoidal waveform. *Physics of Fluids*, 37(11). (IF 4.3, Q1)
28. Li, J., Sun, M., **Zhao, Z.***, & Huang, G. Q. (NOV 2025). Cyber-Physical Internet Spatial-temporal Reasoning for Production Logistics Resource Recommendation in Industry 4.0. *Journal of Industrial Information Integration*, 100935. (IF 11.6, Q1)
27. Xu, Y., Li, S., Feng, K., Huang, R., Sun, B., Yang, X., **Zhao, Z.***, & Huang, G. Q. (JUN 2025). Domain constrained cascadic multireceptive learning networks for machine health monitoring in complex manufacturing systems. *Journal of Manufacturing Systems*, 80, 563–577. (IF: 14.2, Q1)
26. Zhang, M., Yang, W., Wang, P. P., **Zhao, Z.***, Wang, S., & Huang, G. Q. (APR 2025). Towards low-carbon e-commerce return logistics: optimization for simultaneous pickup and delivery. *Industrial Management & Data Systems*. (IF: 4.7, Q2)
25. Wu, W., Zhou, D., Shen, L., **Zhao, Z.***, Li, C., & Huang, G. (JAN 2025). TransAoA: Transformer-based Angle of Arrival Estimation for BLE Indoor Localization. *IEEE Transactions on Instrumentation and Measurement*. (IF: 5.9, Q1)
24. Xu, Y., Li, S., Feng, K., Sun, B., Yang, X., Kou, L., **Zhao, Z.*** & Huang, G. Q., (MAR 2025). Multiperspective Temporal Pooling Convolutional Neural Networks for Fault Diagnosis of Mechanical Transmission Systems. *IEEE Transactions on Instrumentation and Measurement*. (IF: 5.9, Q1)
23. Zhang, M., Shen, Q., **Zhao, Z.***, Wang, S., & Huang, G. Q. (MAR 2025). Optimizing ESG reporting: Innovating with E-BERT models in nature language processing. *Expert Systems with Applications*, 265, 125931. (IF: 7.5, Q1)
22. Sun, M., Ding, J., **Zhao, Z.***, Chen, J., Huang, G. Q., & Wang, L. (FEB 2025). Out-of-order execution enabled deep reinforcement learning for dynamic additive manufacturing scheduling. *Robotics and Computer-Integrated Manufacturing*, 91, 102841. (IF 11.4, Q1)
21. Sun, K., **Zhao, Z.***, Li, M., & Huang, G. Q. (MAR 2025). Multi-order attributes information fusion via hypergraph matching for popular fashion compatibility analysis. *Expert Systems with Applications*, 263, 125758. (IF: 7.5, Q1)
20. Xu, S. X.[#], **Zhao, Z.***, Huang, G. Q., Ding, Y., Li, M., & Feng, J. (JAN 2025). A meta-auction for on-demand transportation procurement in industry 5.0. *Transportation Research Part E: Logistics and Transportation Review*, 193, 103842. ([#]Co-first Author) (IF: 8.8, Q1)
19. **Zhao, Z.***, Xie, X., Chen, Q., Lin, P., Wu, W., Zhang, M., & Huang, G. Q. (DEC 2024). Graduation-inspired manufacturing system and synchronization mechanism for hybrid assembly cell lines. *Computers & Industrial Engineering*, 198, 110648. (IF: 6.5, Q1)
18. Li, P., Wu, W., **Zhao, Z.***, & Huang, G. Q. (DEC 2024). Indoor Positioning Systems in Industry 4.0 Applications: Current Status, Opportunities, and Future Trends. *Digital Engineering*, 100020.
17. Xu, Y., Shu, R., Li, S., Feng, K., Yang, X., **Zhao, Z.***, & Huang, G. Q. (AUG 2024). Imbalanced learning for gearbox fault detection via attention-based multireceptive field convolutional neural networks with an adaptive label regulation loss. *IEEE Transactions on Instrumentation and Measurement*. (IF: 5.9, Q1)
16. Shu, R., Xu, Y., He, J., Yang, X., **Zhao, Z.***, & Huang, G. Q. (OCT, 2024). Multi-view contrastive learning framework for tool wear detection with insufficient annotated data. *Advanced Engineering Informatics*, 62, 102666. (IF: 9.9, Q1)
15. **Zhao, Z.***, Zhang, M., Wu, W., Huang, G. Q., & Wang, L. (JUN 2024). Spatial-temporal traceability for cyber-physical industry 4.0 systems. *Journal of Manufacturing Systems*, 74, 16–29. (IF: 14.2, Q1)

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14. Xu, Y., Jiang, Q., Li, S., **Zhao, Z.***, Sun, B., & Huang, G. Q. (JUL 2024). Digital twin-driven discriminative graph learning networks for cross-domain bearing fault recognition. *Computers & Industrial Engineering*, 110292. (IF: 6.5, Q1)
13. Zhang, M., Shen, Q., **Zhao, Z.***, Wang, S., & Huang, G. Q. (JUN 2024). Risk-averse behavior and incentive policies: A new perspective on spatial-temporal traceability supervision in construction logistics supply chains. *Computers & Industrial Engineering*, 110256. (IF: 6.5, Q1)
12. Zhang, M., Shen, Q., **Zhao, Z.***, Wang, S., & Huang, G. Q. (FEB 2024). Commitment or rent-seeking? Government incentive policies for ESG reporting in sustainable e-commerce logistics. *International Journal of Production Economics*, 268, 109134. (IF: 10, Q1)
11. Zhang, M., Chen, A., **Zhao, Z.***, & Huang, G. Q. (JAN 2024). A multi-depot pollution routing problem with time windows in e-commerce logistics coordination. *Industrial Management & Data Systems*, 124(1), 85-119. (IF: 4.7, Q2)
10. Zhang, M., Yang, W., **Zhao, Z.***, Pratap, S., Wu, W., & Huang, G. Q. (DEC 2023). Is digital twin a better solution to improve ESG evaluation for vaccine logistics supply chain: an evolutionary game analysis. *Operations Management Research*, 1-23. (IF 5.3, Q1)
9. **Zhao, Z.**, Zhang, M., Chen, J., Qu, T., & Huang, G. Q. (SEP 2022). Digital Twin-enabled Dynamic Spatial-temporal Knowledge Graph for Production Logistics Resource Allocation. *Computers & Industrial Engineering*, 108454. (IF: 6.5, Q1)
8. Wu, W., Shen, L., **Zhao, Z.***, Li, M., & Huang, G. Q. (NOV 2022). Industrial IoT and Long Short-Term Memory Network Enabled Genetic Indoor Tracking for Factory Logistics. *IEEE Transactions on Industrial Informatics*. (IF: 9.9, Q1)
7. Yang, C., Lan, S., **Zhao, Z.***, Zhang, M., Wu, W., & Huang, G. Q. (FEB 2023). Edge-cloud blockchain and IoE-enabled quality management platform for perishable supply chain logistics. *IEEE Internet of Things Journal*, 10(4), 3264-3275. (IF: 8.9, Q1)
6. **Zhao, Z.**, Zhong, R. Y., Kuo, Y. H., Fu, Y., & Huang, G. Q. (APR 2021). Cyber-physical spatial temporal analytics for digital twin-enabled smart contact tracing. *Industrial Management & Data Systems*, 121(5), 1082-1106. (IF: 4.7, Q2)
5. **Zhao, Z.**, Shen, L., Yang, C., Wu, W., Zhang, M., & Huang, G. Q. (APR 2021). IoT and digital twin enabled smart tracking for safety management. *Computers & Operations Research*, 128, 105183. (IF: 4.3, Q1)
4. **Zhao, Z.**, Zhang, M., Xu, G., Zhang, D., & Huang, G. Q. (DEC 2020). Logistics sustainability practices: an IoT-enabled smart indoor parking system for industrial hazardous chemical vehicles. *International Journal of Production Research*, 58(24), 7490-7506. (IF: 7.3, Q1)
3. **Zhao, Z.**, Lin, P., Shen, L., Zhang, M., & Huang, G. Q. (JAN 2020). IoT edge computing-enabled collaborative tracking system for manufacturing resources in industrial park. *Advanced Engineering Informatics*, 43, 101044. (IF: 9.9, Q1)
2. **Zhao, Z.**, Zhang, M., Yang, C., Fang, J., & Huang, G. Q. (NOV 2018). Distributed and collaborative proactive tandem location tracking of vehicle products for warehouse operations. *Computers & Industrial Engineering*, 125, 637-648. (IF: 6.5, Q1)
1. **Zhao, Z.**, Fang, J., Huang, G. Q., & Zhang, M. (APR 2017). Location management of cloud forklifts in finished product warehouse. *International Journal of Intelligent Systems*, 32(4), 342-370. (IF: 3.7, Q2)

B.2.2. Refereed journal papers – listed as co-author:

23. Zhu, J., Huang, S., Wang, P., Xu, Z., Liu, J., Wang, B., Zhao, Z., Zheng, S., Tao, Y., Wang, G., Yan, Y., Huang, G. Q., & Wang, L. (2026). A review on large language models for industrial embodied intelligence. *Advanced Engineering Informatics*, 73, 104602.
22. Wang, Y., Cheng, Y., Qi, Q., Zhao, Z., Huang, G. Q., Pickl, S., & Tao, F. (2026). Causality and Equipment Structure Enhanced Maintenance Plan Recommendation with Knowledge Graph Integration. *Engineering*.
21. Chen, J., Ma, W., Sun, M., & Zhao, Z. (2026). Logic-based Benders decomposition for additive manufacturing scheduling on unrelated parallel machines. *Annals of Operations Research*, 1-30.
20. Su, Z., Niu, Y., Li, J., Zhou, Y., Qiu, Y., & Zhao, Z. (NOV 2025). Decentralised schedule for mobile robots in Industry 4.0: an adaptive path planning strategy selection mechanism under dynamic environments. *International Journal of Production Research*, 1-20. (IF: 7.3, Q1)
19. Zhang, M., He, S., Wang, P. P., **Zhao, Z.**, Wang, S., & Huang, G. Q. (JUN, 2025). Channel structures and ESG initiatives for logistics service providers. *Transport Policy*, 166, 18-31. (IF: 5.3, Q1)
18. Sun, K., Zhao, Z., Yang, H., Zhang, J., & Huang, G. Q. (OCT 2025). Curriculum Engineering: Structured Learning for Large Language Models (LLMs) Through Curriculum Based Retrieval. *IEEE Transactions on Industrial Informatics*. (IF: 9.9, Q1)

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17. Chen, Q., **Zhao, Z.**, Ouyang, Z., Wang, Y. J., Yuan, Z., & Huang, G. Q. (SEP 2025). Cyber-physical internet shipment execution protocols for aviation maintenance, repair and overhaul intralogistics. *Computers & Industrial Engineering*, 111276. (IF: 6.5, Q1)
16. Yu, S., Zhang, M., **Zhao, Z.**, Wang, P. P., & Huang, G. Q. (JUN 2025). ESG transformation through private equity and digital twin in energy supply chain: an evolutionary game analysis. *Industrial Management & Data Systems*. (IF: 4.7, Q2)
15. Li, Z., Yan, Y., **Zhao, Z.**, Xu, Y., & Hu, Y. (OCT 2025). Numerical study on hydrodynamic effects of intermittent or sinusoidal coordination of pectoral fins to achieve spontaneous nose-up pitching behavior in dolphins. *Ocean Engineering*, 337, 121854. (IF: 5.5, Q1)
14. Ouyang, Z., Yuan, Z., Chen, Q., Li, M., **Zhao, Z.**, & Huang, G. Q. (OCT 2025). Shipment Scheduling and Routing Protocols in Cyber-Physical Internet for Prefabricated Construction Modules Logistics. *IEEE Transactions on Intelligent Transportation Systems*. (IF: 8.4, Q1)
13. Wu, H., Li, M., Yu, C., Ouyang, Z., Lai, K. H., **Zhao, Z.**, ... & Huang, G. Q. (MAY 2025). Towards cyber-physical internet: A systematic review, fundamental model and future perspectives. *Transportation Research Part E: Logistics and Transportation Review*, 197, 104051. (IF: 8.8, Q1)
12. Li, S., Jiang, Q., Xu, Y., Feng, K., **Zhao, Z.**, Sun, B., & Huang, G. Q. (OCT 2024). Digital twin-assisted interpretable transfer learning: A novel wavelet-based framework for intelligent fault diagnostics from simulated domain to real industrial domain. *Advanced Engineering Informatics*, 62, 102681. (IF: 9.9, Q1)
11. Tan, B. Q., Xu, S. X., Thüerer, M., Kang, K., **Zhao, Z.**, & Li, M. (MAY 2024). Booking versus search-based parking strategy: A game-theoretic methodology. *Research in Transportation Economics*, 104, 101416. (IF: 3.4, Q1)
10. Chen, J., Ma, W., Ye, X., & **Zhao, Z.** (OCT 2023). Logic-based Benders decomposition for order acceptance and scheduling in distributed manufacturing. *Advanced Engineering Informatics*, 58, 102228. (IF: 9.9, Q1)
9. Wu, W., Shen, L., **Zhao, Z.**, Harish, A. R., Zhong, R. Y., & Huang, G. Q. (JUN 2023). Internet of Everything and Digital Twin Enabled Service Platform for Cold Chain Logistics. *Journal of Industrial Information Integration*, 33, 100443. (IF: 11.6, Q1)
8. Zhan, X., Wu, W., Shen, L., Liao, W., **Zhao, Z.**, & Xia, J. (AUG 2022). Industrial internet of things and unsupervised deep learning enabled real-time occupational safety monitoring in cold storage warehouse. *Safety Science*, 152, 105766. (IF: 5.4, Q1)
7. Wu, W., **Zhao, Z.**, Shen, L., Kong, X. T., Guo, D., Zhong, R. Y., & Huang, G. Q. (APR 2022). Just Trolley: Implementation of industrial IoT and digital twin-enabled spatial-temporal traceability and visibility for finished goods logistics. *Advanced Engineering Informatics*, 52, 101571. (IF: 9.9, Q1)
6. Zhang, M., Pratap, S., **Zhao, Z.**, Prajapati, D., & Huang, G. Q. (OCT 2021). Forward and reverse logistics vehicle routing problems with time horizons in B2C e-commerce logistics. *International Journal of Production Research*, 59(20), 6291-6310. (IF: 7.3, Q1)
5. Kong, X. T., Zhong, R. Y., **Zhao, Z.**, Shao, S., Li, M., Lin, P., ... & Huang, G. Q. (JAN, 2020). Cyber physical ecommerce logistics system: An implementation case in Hong Kong. *Computers & Industrial Engineering*, 139, 106170. (IF: 6.5, Q1)
4. Lin, P., Shen, L., **Zhao, Z.**, & Huang, G. Q. (DEC 2019). Graduation manufacturing system: synchronization with IoT-enabled smart tickets. *Journal of Intelligent Manufacturing*, 30(8), 2885-2900. (IF: 7.4, Q1)
3. Zhang, M., Huang, G. Q., Xu, S. X., & **Zhao, Z.** (OCT 2019). Optimization based transportation service trading in B2B e-commerce logistics. *Journal of Intelligent Manufacturing*, 30(7), 2603-2619. (IF: 7.4, Q1)
2. Zhang, M., Fu, Y., **Zhao, Z.**, Pratap, S., & Huang, G. Q. (MAR 2019). Game theoretic analysis of horizontal carrier coordination with revenue sharing in E-commerce logistics. *International Journal of Production Research*, 57(5), 1524-1551. (IF 7.3 Q1)
1. Zhang, M., Pratap, S., Huang, G. Q., & **Zhao, Z.** (APR 2017). Optimal collaborative transportation service trading in B2B e-commerce logistics. *International Journal of Production Research*, 55(18), 5485-5501. (IF: 7.3, Q1)

B.2.3. Conference papers:

22. Yue, P., Liu, H., Guo, W., Shen, L., Li, P., **Zhao, Z.**, & Huang, G. Q. (2025, September). A LiDAR-guided transformer framework for low-cost Bluetooth AOA positioning with adaptive Kalman filtering. In *IET Conference Proceedings CP942* (Vol. 2025, No. 28, pp. 28-34). Stevenage, UK: The Institution of Engineering and Technology.
21. Han, Y., Sun, K., **Zhao, Z.**, & Huang, G. Q. (2025, August). Out-of-Distribution Modular Hospital Fit-Out Scheduling via Memory-augmented Deep Reinforcement Learning. In *2025 IEEE 21st International Conference on Automation Science and Engineering (CASE)* (pp. 1292-1297). IEEE.

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20. Sun, K., Han, Y., **Zhao, Z.**, & Huang, G. Q. (2025, August). CL-Tuning: A Curriculum Learning-Based Progressive Fine-Tuning Framework for Large Language Models in Social Manufacturing. In *2025 IEEE 21st International Conference on Automation Science and Engineering (CASE)* (pp. 3373-3378). IEEE.
19. Huang, Y., Lam, H. S., **Zhao, Z.**, & Huang, G. Q. (2025, August). UNIQC: Unified I/Q Correction with Quality-Scored Filtering for BLE-Enabled Application Traceability for Industry 4.0 Applications. In *2025 IEEE 21st International Conference on Automation Science and Engineering (CASE)* (pp. 2301-2306). IEEE.
18. Sun, M., Li, J., Chen, Q., Zhang, M., **Zhao, Z.**, & Huang, G. Q. (2025, August). Dynamic Double-sided Rolling-Horizon Auction Mechanisms for Additive Manufacturing Collaboration in Social Manufacturing. In *2025 IEEE 21st International Conference on Automation Science and Engineering (CASE)* (pp. 631-636). IEEE.
17. Li, P., Liu, H., Guo, W., Yue, P., Shen, L., **Zhao, Z.**, & Huang, G. Q. (2025, August). Transfer Learning-driven Scalability for Indoor Positioning Systems in Industrial Environments. In *2025 IEEE 21st International Conference on Automation Science and Engineering (CASE)* (pp. 941-946). IEEE.
16. Wang, Y., **Zhao, Z.**, & Huang, G. Q. (2025). Learning Towards Fair Order Dispatching via Hierarchical Attention-based Reinforcement Learning for Garment Manufacturing. *IFAC-PapersOnLine*, 59(10), 2070-2075.
15. Li, J., Sun, M., **Zhao, Z.**, & Huang, G. Q. (2025). Production Logistics Resource Recommendation Based on ‘Look-Around’ Reasoning Mechanism in Discrete Manufacturing. *IFAC-PapersOnLine*, 59(10), 2052-2057.
14. Chen, Q., **Zhao, Z.**, Luo, H., & Huang, G. G. (2023, August). IoT-Enabled MRO Intralogistics Management: A Case Study in Hong Kong. In *2023 IEEE 19th International Conference on Automation Science and Engineering (CASE)* (pp. 1-6). IEEE.
13. Kong, X. T., Li, M., Yu, Y., **Zhao, Z.**, & Huang, G. Q. (2017, August). Physical internet-enabled E-commerce logistics park platform. In *2017 13th IEEE Conference on Automation Science and Engineering (CASE)* (pp. 406-411). IEEE.
12. Zhou, D., **Zhao, Z.**, Wu, W., Zhang, M., Xu, G., Zhang, M., & Huang, G. Q. (2024, October). Empowering Bluetooth Angle of Arrival Positioning with Ultra-wideband for Industry 4.0. In *CEUR Workshop Proceedings* (Vol. 3919, p. ecopy). CEUR-WS.
11. **Zhao, Z.**, Fang, J., Huang, G. Q., & Zhang, M. (2016, September). iBeacon enabled indoor positioning for warehouse management. In *Computational and Business Intelligence (ISCBI), 2016 4th International Symposium on* (pp. 21-26). IEEE.
10. Chen, Q., Ding, J., Sun, M., **Zhao, Z.**, & Huang, G. Q. (2024, September). A Dynamic Fit-Out Scheduling Framework for Digital Twin-Enabled Modular Integrated Construction. In *IFIP International Conference on Advances in Production Management Systems* (pp. 168-179). Cham: Springer Nature Switzerland.
9. Sun, M., Ding, J., Chen, J., **Zhao, Z.**, & Huang, G. Q. (2024, August). Optimization of Dynamic Scheduling in Additive Manufacturing with Deep Reinforcement Learning. In *2024 IEEE 20th International Conference on Automation Science and Engineering (CASE)* (pp. 3352-3357). IEEE.
8. Chen, Q., **Zhao, Z.**, & Huang, G. Q. (2024, August). Protocol Model and Execution Protocols for Networked MRO Intralogistics. In *2024 IEEE 20th International Conference on Automation Science and Engineering (CASE)* (pp. 2553-2558). IEEE.
7. Yuan, Z., Ding, J., Li, M., Huang, G. Q., & **Zhao, Z.** (2024, August). Solving Size-Agnostic Job Shop Scheduling Problems Like GPT Speaks. In *2024 IEEE 20th International Conference on Automation Science and Engineering (CASE)* (pp. 3346-3351). IEEE.
6. Li, P., Xie, J., Ding, J., **Zhao, Z.**, Wu, W., & Huang, G. Q. (2024, August). Real-time Locating System-enabled Digital Twin for Crane Operation Safety Monitoring on Construction Sites. In *2024 IEEE 20th International Conference on Automation Science and Engineering (CASE)* (pp. 2689-2696). IEEE.
5. Han, Y., Xie, J., Ding, J., **Zhao, Z.**, & Huang, G. Q. (2024, August). DT-PoseFormer: A Digital Twin-enabled Stacking System for Precise Pose Estimation of MiC Modules. In *2024 IEEE 20th International Conference on Automation Science and Engineering (CASE)* (pp. 2678-2683). IEEE.
4. Yang, H., **Zhao, Z.**, & Zhang, M. (2023, November). Blockchain and SC-PBFT Enabled Traceability Method for Vaccine Supply Chain. In *2023 China Automation Congress (CAC)* (pp. 3639-3645). IEEE.
3. Chen, Q., **Zhao, Z.**, & Huang, G. Q. (2023, October). Cyber-Physical Internet Local Area Network for Aviation Maintenance, Repair and Overhaul (mro) Intralogistics. In *50th International Conference on Computers and Industrial Engineering: Sustainable Digital Transformation, CIE 2023* (pp. 981-990). Computers and Industrial Engineering.
2. Kong, X. T., Li, M., Yu, Y., **Zhao, Z.**, & Huang, G. Q. (2017, August). Physical internet-enabled E-commerce logistics park platform. In *2017 13th IEEE conference on automation science and engineering (CASE)* (pp. 406-411). IEEE.

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1. Zhao, Z., Fang, J., Huang, G. Q., & Zhang, M. (2016, September). iBeacon enabled indoor positioning for warehouse management. In *2016 4th international symposium on computational and business intelligence (ISCBI)* (pp. 21-26). IEEE.

B.2.4. Patents (Invention)

1. Zhiheng Zhao, Aoxiang Chen, Mengdi Zhang, Dengyin Zhang. 'A production logistics resource allocation method based on dynamic spatiotemporal knowledge graph'. ZL 202210298702.7 (Granted)
2. Zhiheng Zhao, Mengdi Zhang, Dengyin Zhang. 'A method and system for identifying and locating a trailer used in a hazardous chemical parking lot'. ZL201911070277.0 (Granted and Transferred)
3. Zhiheng Zhao, Huixing Yang, Dengyin Zhang. 'A cold chain logistics quality management system'. CN202210748327.1(In examination)
4. Zhiheng Zhao, Mengdi Zhang, Dengyin Zhang, Shasha Zhao. 'An indoor and outdoor joint positioning method and device for a work vehicle based on Internet of Things'. CN202010088014.9 (In examination)
5. Zhiheng Zhao, Qing Zhong. 'A smart warehouse management system'. CN201910346371.8 (In examination)

B.2.5. Patents (Utility model)

1. Guoquan Huang, Ji Fang, Zhiheng Zhao. 'A distributed multi-functional indoor positioning base station'. CN201420583450.3 (Granted)
2. Ji Fang, Guoquan Huang, Zhiheng Zhao. 'A smart lamp holder with Bluetooth positioning function'. CN201520108852.2 (Granted)
3. Ming Li, Zhiheng Zhao, Leidi Shen, 'An order picking system', CN215754511U, 08/02/2022. (Granted)
4. Ming Li, Zhiheng Zhao, Leidi Shen, 'An inventory system', CN215006710U, 03/12/2021. (Granted)

B.2.6. Software Copyrights

1. 'CPS Additive reality based inbound management mobile application system' V1.0 2020SR0521601.
2. 'Smart logistics buffering management system' V1.1, 2019SR0801270.
3. 'Smart warehouse buffering management system' V1.1, 2019SR0800812.

C DETAILS OF TEACHING ACHIEVEMENTS

Billy has taught five courses at PolyU, employing the TELD (Teaching by Examples and Learning by Doing) pedagogy. This approach has proven highly effective, yielding eSFQ scores of up to 4.9 (above the department average of 4.3). He co-leads curriculum development for the Master of Smart Manufacturing program and introduced a new postgraduate course. Billy currently supervises or co-supervises 16 research students and 15 research staff.

C.1. TEACHING ACTIVITIES AND EVALUATION

	Course Name/Code	Level	Semester	Evaluation
Course taught at PolyU	Integrated Design for Manufacture, ISE386	UG	2025/26 S1	eSFQ 4.3/5.0
	Smart manufacturing operations management, ISE5332	PG	2024/25 S2	eSFQ 4.9/5.0
	Integrated Design for Manufacture, ISE386	UG	2024/25 S1	eSFQ 4.3/5.0
	Project Management, ENG4001	UG	2023/24 S2	N.A.
	Integrated Design for Manufacture, ISE386	UG	2023/24 S1	eSFQ 4.6/5.0
	Data Management in Aviation Industries, ISE3013	UG	2023/24 S1	N.A.
Course taught before joining PolyU	Computer Networks, B1801381S	UG	2020/21 S1	Ranking 1/34
	MATLAB Simulation, B1802021C	UG	2020/21 S1	N.A.
	Introduction to IoT, B18010K2C	UG	2019/20 S2	Ranking 8/28
	Computer Networks, K1801091S	UG	2019/20 S1	N.A.
	Introduction to IoT, B18010K2C	UG	2018/19 S2	Ranking 1/31

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C.2. SUPERVISION OF RESEARCH STUDENTS/ASSISTANTS/ASSOCIATES/PDFS

of Ph.D. (MPhil) under supervision: 14 (2)

of Research Associate/PDF under supervision: 10

of Research/Project Assistant under supervision: 5

PhD students under supervision:

Chief supervised: Mr. LIU Haoran(2025), Ms. LIU Qi(2025), Mr. CHEN Junyang(2025), Mr. LI Jinpeng (2024), Ms. HAN Yujie(2024), Mr. LI Peisen(2024),

Co-supervised Ms. XIE Xinyue (2025), Mr. YUE Pengjun (2025), Ms. SUN Kexin (2025), Mr. CHENG Yuan (2025), Mr. MA Siyuan (2025), Mr. YU Shengan (2025), Ms. SUN Mingyue (2024), Ms. WANG Yanying (2024)

MPhil students under supervision:

Co-supervised: Mr. HUANG Bin (2024), Ms. SONG Wanyi (2024)

Research Associate/PDF:

Dr. WANG Ming (2025), Dr YUAN Yue (2025), Dr. LI Zhihan (2025), Dr. LU Zefeng (2025), Dr. GUO Wei (2024), Dr. SHEN Leidi (2024), Dr. WANG Yijia (2025), Dr. XU Yadong (2024), Dr. FENG Yongbing (2025), Dr. SU Ruoyu (2025), Dr. CHEN Huanhui (2025)

Research/Project Assistant:

Ms. HUANG Yijia (2024), Ms. LIN Jialing (2024) Mr. YING Binbin (2025), Ms. SUN Hui(2025), Ms. ZHANG Fan (2025)

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D DETAILS ON PROFESSIONAL SERVICES AND ENTREPRENEURSHIP

Within the international community, Billy has been very active in professional activities and contributions. As program chairs, he has recently organized 11th International Physical Internet Conference, with more than two hundred attendees. He has served as the Co-founder and CEO of Gantan Limited since 2016, backed by the investments from Cyberport and Hangzhou Municipal Government. Major clients include flagship manufacturing and logistics leaders such as Huawei(華為), Lenovo(聯想), Gree Electric Appliances. Inc(格力電器), DKSH (大昌華嘉), HACTL(香港空運貨站).

D.1. ACADEMIC SERVICES

Associate Editor	<i>International Journal of Manufacturing Research</i>
Leading Guest Editor	Special Issue ‘Smart Logistics and Supply Chain Management with AI-empowered Physical Internet’ in <i>International Journal of Logistics Research and Applications</i>
Program Chair	11 th International Physical Internet Conference, 18-20 June 2025, Hong Kong. https://ipic2025.pi.events/
Program Chair	4 th IEEE International Conference on Automation in Manufacturing, Transportation and Logistics (iCaMaL2024), 7-9 Aug 2024, Hong Kong.
Organizing Chair	2025 大湾区物流与供应链创新联盟年会及研讨会：人工智能+赋能供应链与物流管理创新, 31 October – 2 November 2025, Zhuhai, China.
Organizing Chair	Conference on Intelligent Networked Things (CINT2025), 13-25 June 2025, Zhuhai, China
Organizing Chair	2024 Annual International Conference for Chinese Scholars in Industrial Engineering & 14 th Forum for Council of Industrial Engineering and Logistics Management Department Heads, 9-11 Aug 2024, Hong Kong.
Organizing Chair	The 19 th International Congress on Logistics and SCM Systems, 7-9 June 2025, Hong Kong
Special Session Chair	IEEE CASE 2024 Digital twin in smart construction
Special Session Chair	IEEE CASE 2024 Blockchain-empowered industrial web 3.0
Special Session Chair	IEEE CASE 2024 Cyber-physical manufacturing for small batch customization: challenges, enabling technologies, and methodologies
Special Session Chair	IEEE SMC 2024 Smart and sustainable manufacturing systems
Publication and Award Member	DILSCM 2025, Guangzhou, China
Reviewer	<i>Nature Communications, IEEE Transactions on Wireless Communications, IEEE Internet of Things Journal, Journal of Manufacturing Systems, IEEE Transactions on Engineering Management, International Journal of Digital Earth, Digital Twin, Advanced Engineering Informatics, Automation in Construction, Computers & Industrial Engineering, Computers & Operations Research, International Journal of Computer Integrated Manufacturing, International Journal of Production Research, Transportation Research Part E, Annals of Operations Research, Journal of Advanced Transportation, International Journal of Production Economics, Journal of Industrial Information Integration, Journal of Network and Computer Applications., etc</i>

D.2. INVITED TALKS

1. 應用人工智能技術實現慢性腎臟病的早期檢測, November 2025, People’s Liberation Army General Hospital and Medical School (301 Hospital), Beijing, China.
2. From Ideas to Impact: My Journey in Innovation and Entrepreneurship, G7 Students Workshop, May 2025, Canadian International School of Hong Kong, Hong Kong SAR
3. Industry 4.0 Smart Prefabrication Yard for Modular Construction Fitout, October 2024, Northeast Agricultural University, Harbin, China.
4. Spatial-temporal Traceability for Cyber-physical Industry 4.0 Systems, Vision and AI forum, Apr 2024, Beihang University, Beijing, China.
5. Spatial-temporal Traceability for Cyber-physical Industry 4.0 Systems, IEEE Technical Committee on Digital Manufacturing and Human-Centric Automation Workshop, Jan 2024, The Hong Kong Polytechnic University, Hong Kong SAR.
6. On Spatial-temporal Traceability and Relativity for Industry 4.0 Applications, Annual International Conference for Digital Intelligence Logistics and Supply Chain Management, Nov 2023, Shenzhen Technology University,

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Shenzhen China.

7. SynchronHub: Cyber-Physical Internet for Synchronizing Cross-Border Logistics Hubs in the Greater Bay Area (GBA), August 2023, 2nd PAIR Salon, Interdisciplinary Dialogue, Science Park, Hong Kong SAR.

D.3. ENTREPRENEURSHIP

Company Name	Year of Registration	Position	URL	
Gantan Limited	2016	Co-founder/CEO	www.gantanmark.com	
No.	Program/Project Title	Funding Source	Amount of Investments / Project	Type
1	Cyberport Incubation Program	<i>Hong Kong Cyberport Management Company Limited</i>	HK\$330,000	Investments
2	Hangzhou Qingshan Lake Innovation and Entrepreneurship Project	<i>Hangzhou Government</i>	CNY ¥ 4,000,000	
3	IoT assisted indoor positioning for air cargo logistics management	<i>Hong Kong Air Cargo Terminals Ltd</i>	HK\$2,465,840	Major contracts
4	Bluetooth Low Energy enabled indoor positioning for material trolleys	<i>Lenovo Group Ltd</i>	CNY ¥ 274,800	
5	IoT tracking for cold-chain logistics COVID-19 vaccine quality management	<i>DKSH Management Ltd</i>	CHF\$40,000	
6	Manufacturing resource tracking system	<i>Gree Electric Appliances. Inc</i>	CNY ¥ 90,000	
7	Cloud asset tracking management platform	<i>Huawei Technologies Ltd</i>	US\$30,000	
8	Supply chain logistics platform for smart furniture	<i>Redstar Mancalline Co. Ltd</i>	HK\$900,000	
9	IoT based industry vehicles tractor-trailer positioning	<i>Zhejiang Juhua Group Corp</i>	CNY ¥ 477,000	

D.4. PROFESSIONAL MEMBERSHIPS

Senior Member, IEEE	The Institute of Electrical and Electronics Engineers (IEEE), USA
Member, HKIE	Hong Kong Institution of Engineers (HKIE), Hong Kong
Member, ASME	The American Society of Mechanical Engineers (ASME), USA
Member, CCF	China Computer Federation (CCF), China
Member, IISE	Institute of Industrial and Systems Engineers (IISE), USA
Member, CMES	China Mechanical Engineering Society (CMES), China
Member, CNS	China Nuclear Society (CNS), China

D.5. SERVICE FOR PROFESSIONAL SOCIETIES AND COMMUNITIES

Council member	Guangdong Mechanical Engineering Society
Council member	粵港澳大灣區物流與供應鏈創新聯盟學術研究委員會
Council member	Logistics Branch of Guangdong Operations Research Society
Committee member	Digital Twin Young Scholar (China)
Committee member	IEEE Technical Committee on Digital Manufacturing and Human-centred Automation